

**Laboratorium
Multimedia dan Internet of Things
Departemen Teknik Komputer
*Institut Teknologi Sepuluh Nopember***

Laporan Sementara Praktikum Jaringan Komputer

Firewall & NAT

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1 Pendahuluan

1.1 Latar Belakang

Kebutuhan enterprise mengenai internet sangat kompleks dan luas. Namun, hal tersebut dapat didekati melalui pendekatan sederhana yaitu jaringan komputer. Dalam kebutuhan enterprise, salah satu penerapan jaringan komputer ada pada konfigurasi firewall dan Network Address Translation (NAT). Pada praktikum ini, diberikan pengetahuan, wawasan, dan tugas untuk aplikasi dari firewall dan NAT.

1.2 Dasar Teori

1.2.1 Pengenalan Firewall

Firewall bertindak sebagai sistem keamanan yang berdiri di gerbang jaringan untuk memeriksa lalu lintas data yang masuk maupun keluar. Sistem ini mengevaluasi setiap paket data berdasarkan aturan keamanan untuk menentukan kebijakan akses yaitu Accept, Reject, dan Drop. Firewall menutupi kelemahan Access Control List (ACL) tradisional yang tidak bisa menginspeksi isi paket, sehingga jaringan internal terlindungi dari peretas, perangkat perusak, dan celah keamanan eksternal lainnya.

1.2.2 Jenis-Jenis Firewall

Terdapat berbagai varian firewall berdasarkan metode inspeksi dan implementasinya yaitu Packet Filtering, Stateful Inspection, Application Layer Firewall, Next Generation Firewall, Circuit Level Gateway, Software Firewall, Hardware Firewall, dan Cloud Firewall. Packet Filtering mengecek data berdasarkan IP, port, dan protokol secara kaku, sedangkan Stateful Inspection mampu melacak status koneksi yang sah. Teknologi yang lebih maju seperti Next Generation Firewall bahkan mendukung inspeksi mendalam hingga ke isi data, tingkat aplikasi, dan lalu lintas terenkripsi.

1.2.3 Network Address Translation (NAT)

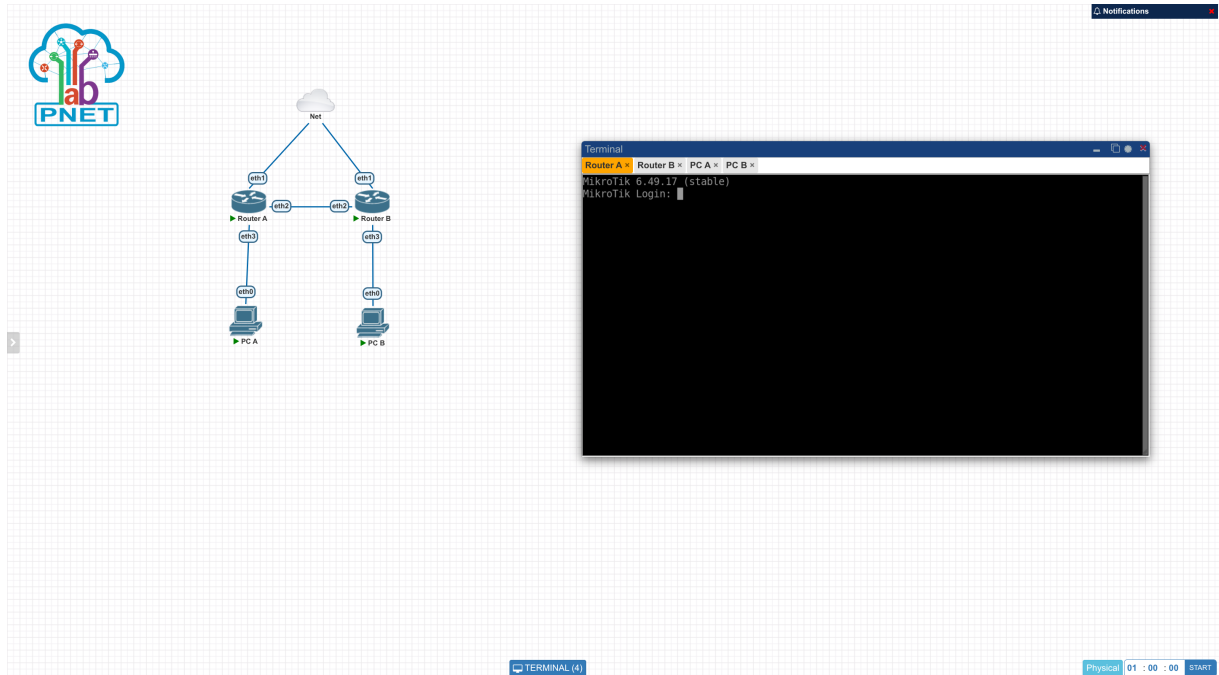
Network Address Translation atau NAT merupakan mekanisme yang memungkinkan banyak perangkat di jaringan lokal untuk mengakses internet menggunakan satu alamat IP publik saja. NAT menyelesaikan masalah keterbatasan IPv4 dengan mentranslasikan alamat jaringan privat (Inside Local Address) ke IP publik (Inside Global Address) saat paket keluar, dan sebaliknya saat paket masuk. Terdapat tiga jenis implementasi NAT yaitu Static NAT, Dynamic NAT, dan Port Address Translation (PAT). PAT merupakan metode yang paling efisien karena membedakan banyak koneksi lokal menggunakan modifikasi nomor port.

1.2.4 Connection Tracking

Connection Tracking adalah modul pengamat jaringan yang memonitor dan mencatat informasi sesi komunikasi aktif yaitu alamat sumber, alamat tujuan, port sumber, port tujuan, protokol, dan status koneksi. Pencatatan ini menjadi fondasi penting bagi proses NAT, dan firewall filtering karena memungkinkan router secara otomatis mengenali paket balasan yang sah, serta menolak koneksi asing. Fitur ini pada akhirnya meningkatkan keamanan firewall secara keseluruhan, mendukung efisiensi NAT, dan mengurangi beban pemrosesan pada router.

2 Tugas Pendahuluan

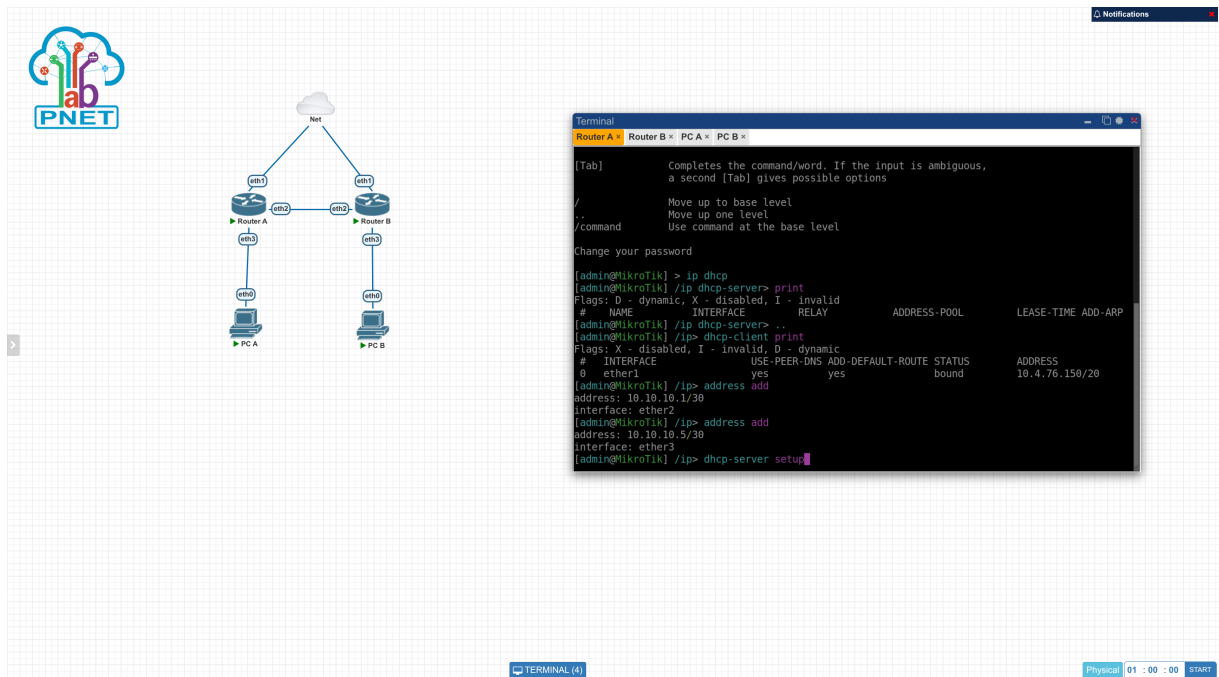
1.



The screenshot shows the PNET network simulator interface. On the left, there is a network topology diagram. It features a central cloud labeled 'Net' connected to two routers, 'Router A' and 'Router B'. Router A is connected to 'eth1' and 'eth2' ports, while Router B is connected to 'eth1' and 'eth2' ports. Both routers have 'eth3' ports connected to two PCs, 'PC A' and 'PC B'. On the right, a terminal window is open, displaying the MikroTik login prompt: 'MikroTik 6.49.17 (stable) MikroTik Login:'. The terminal window has tabs for 'Router A', 'Router B', 'PC A', and 'PC B'. At the bottom of the interface, there is a status bar showing 'TERMINAL (4)' and 'Physical 01 : 00 : 00 START'.

Pertama-tama, rangkai jaringan sesuai skema.

2.



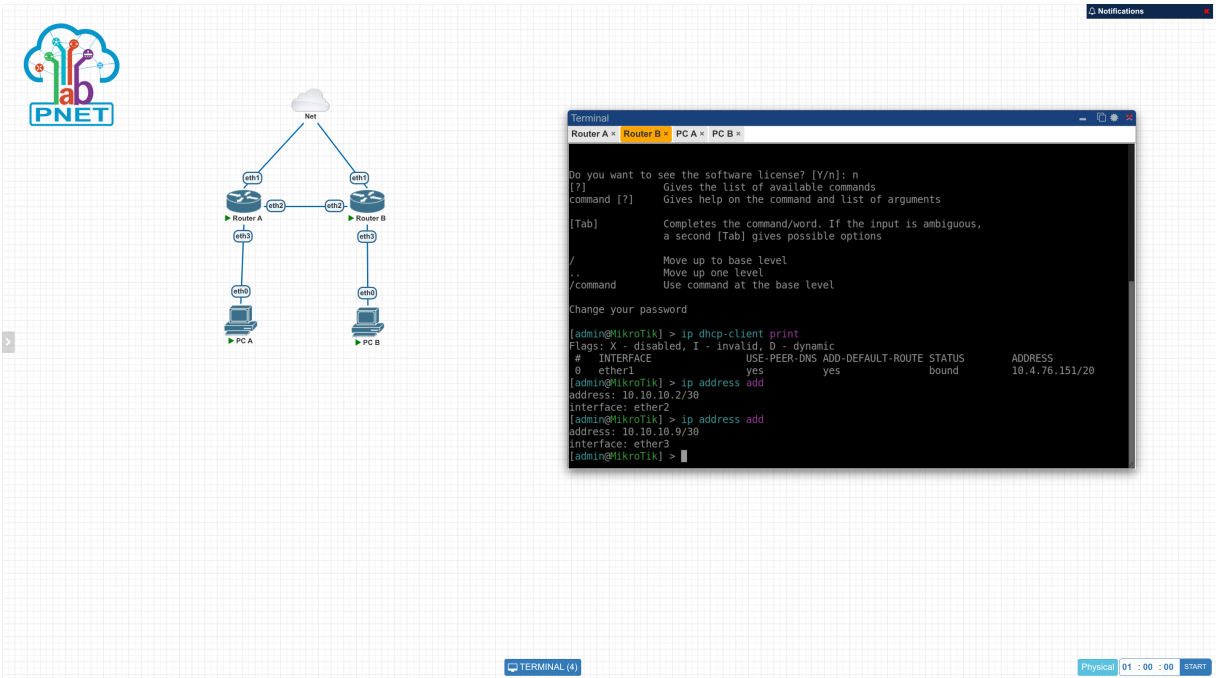
The screenshot shows the PNET network simulator interface with the same network topology as in the first screenshot. On the right, the terminal window is open, displaying the MikroTik login prompt. The user has entered the command 'ip dhcp' and is now in the DHCP configuration menu. The terminal shows the following commands and output:

```
[admin@MikroTik] > ip dhcp
[admin@MikroTik] /ip dhcp-server> print
Flags: D - dynamic, X - disabled, I - invalid
# NAME INTERFACE RELAY ADDRESS-POOL LEASE-TIME ADD-ARP
[admin@MikroTik] /ip dhcp-server> ..
[admin@MikroTik] /ip> dhcp-client print
Flags: X - disabled, I - invalid, D - dynamic
# INTERFACE USE-PEER-DNS ADD-DEFAULT-ROUTE STATUS ADDRESS
0 ether1 yes yes bound 10.4.76.150/20
[admin@MikroTik] /ip> address add
address: 10.10.10.1/30
interface: ether2
[admin@MikroTik] /ip> address add
address: 10.10.10.5/30
interface: ether3
[admin@MikroTik] /ip> dhcp-server setup
```

The terminal window has tabs for 'Router A', 'Router B', 'PC A', and 'PC B'. At the bottom of the interface, there is a status bar showing 'TERMINAL (4)' and 'Physical 01 : 00 : 00 START'.

Kemudian, konfigurasi IP dari setiap port ethernet dan DHCP Management Cloud di Router A

3.



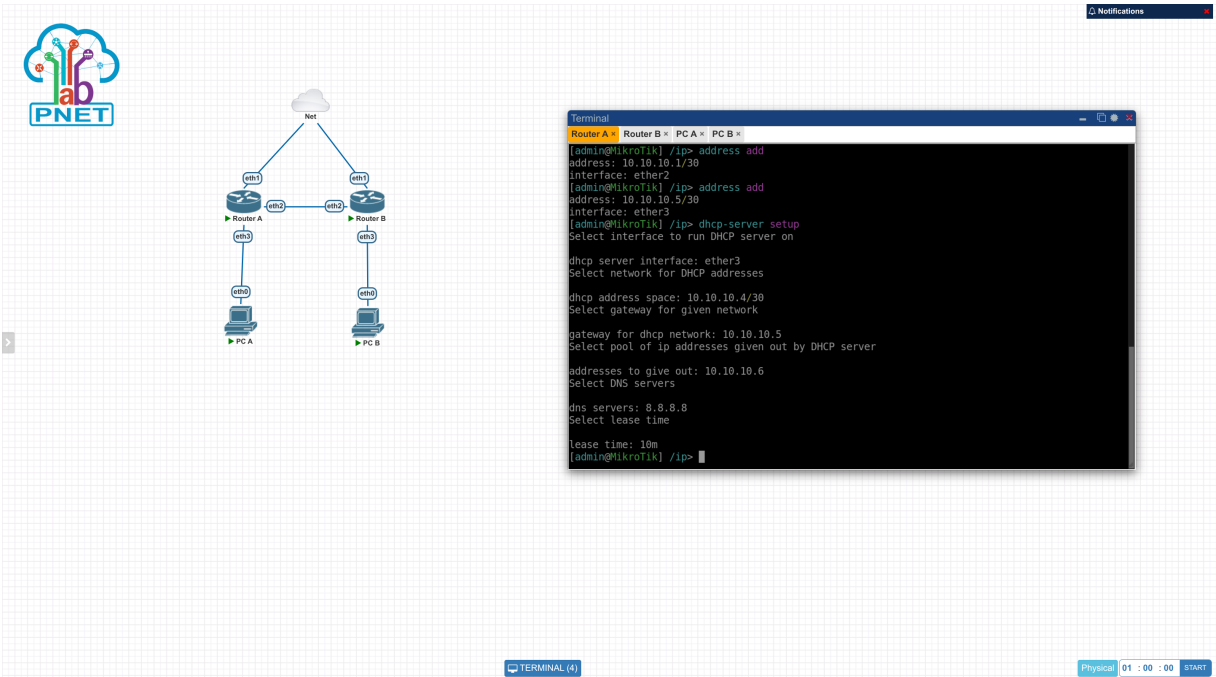
The network diagram shows a central cloud labeled 'Net' connected to two routers, Router A and Router B. Router A is connected to PC A, and Router B is connected to PC B. Both routers have multiple Ethernet ports labeled 'eth0' through 'eth5'. The terminal screenshot shows the MikroTik WinBox interface with the following commands and output:

```

[admin@MikroTik] > ip dhcp-client print
Flags: X - disabled, I - invalid, D - dynamic
# INTERFACE USE-PEER-DNS ADD-DEFAULT-ROUTE STATUS ADDRESS
0 ether1 yes yes bound 10.4.76.151/20
[admin@MikroTik] > ip address add
address: 10.10.10.2/30
interface: ether2
[admin@MikroTik] > ip address add
address: 10.10.10.9/30
interface: ether3
[admin@MikroTik] >
  
```

Dilanjut dengan konfigurasi IP dari setiap port ethernet dan DHCP Management Cloud di Router B

4.



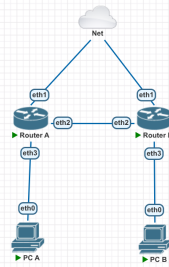
The network diagram is identical to the one in step 3. The terminal screenshot shows the MikroTik WinBox interface with the following commands and output:

```

[admin@MikroTik] /ip> address add
address: 10.10.10.1/30
interface: ether2
[admin@MikroTik] /ip> address add
address: 10.10.10.5/30
interface: ether3
[admin@MikroTik] /ip> dhcp-server setup
Select interface to run DHCP server on
dhcp server interface: ether3
Select network for DHCP addresses
dhcp address space: 10.10.10.4/30
Select gateway for given network
gateway for dhcp network: 10.10.10.5
Select pool of ip addresses given out by DHCP server
addresses to give out: 10.10.10.6
Select DNS servers
dns servers: 8.8.8.8
select lease time
lease time: 10m
[admin@MikroTik] /ip>
  
```

Lakukan konfigurasi DHCP Server di Router A

5.



```

Terminal
Router A * Router B * PCA * PC B *
[admin@MikroTik] > ip address add
address: 10.10.10.2/30
interface: ether2
[admin@MikroTik] > ip address add
address: 10.10.10.9/30
interface: ether3
[admin@MikroTik] > ip dhcp-server setup
Select interface to run DHCP server on
dhcp server interface: ether3
Select network for DHCP addresses
dhcp address space: 10.10.10.8/30
Select gateway for given network
gateway for dhcp network: 10.10.10.9
Select pool of ip addresses given out by DHCP server
addresses to give out: 10.10.10.10
Select DNS servers
dns servers: 8.8.8.8
Select lease time
lease time: 10m
[admin@MikroTik] >

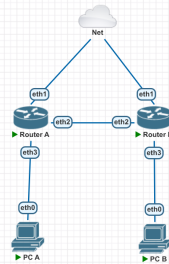
```

TERMINAL (4)

Physical 01 : 00 : 00 START

Lakukan konfigurasi DHCP Server di Router B

6.



```

Terminal
Router A * Router B * PCA * PC B *
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team
Press '?' to get help.
Can't open A
VPCS> show ip
NAME      : VPCS[1]
IP/MASK    : 0.0.0.0/0
GATEWAY    : 0.0.0.0
DNS        :
MAC        : 00:50:79:66:68:7d
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
VTU        : 1500
VPCS> ip dhcp
DORA IP 10.10.10.6/30 GW 10.10.10.5
VPCS> show ip
NAME      : VPCS[1]
IP/MASK    : 10.10.10.6/30
GATEWAY    : 10.10.10.5
DNS        : 8.8.8.8 202.46.129.3
DHCP SERVER : 10.10.10.5
DHCP LEASE  : 598, 600/300/525
MAC        : 00:50:79:66:68:7d
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
VTU        : 1500
VPCS>

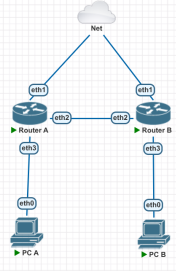
```

TERMINAL (4)

Physical 01 : 00 : 00 START

Kemudian, reservasi IP di PC A dengan berinteraksi dengan DHCP Router A

7.

```

Terminal
Router A * Router B * PCA * PC B *
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.
Can't open B

VPCS> show ip

NAME      : VPCS[1]
IP/MASK    : 0.0.0.0/0
GATEWAY    : 0.0.0.0
DNS        :
MAC        : 00:50:79:66:68:7e
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VPCS> ip dhcp
DORA IP 10.10.10.10/30 GW 10.10.10.9

VPCS> show ip

NAME      : VPCS[1]
IP/MASK    : 10.10.10.10/30
GATEWAY    : 10.10.10.9
DNS        : 8.8.8.8 202.46.129.3
DHCP SERVER : 10.10.10.9
DHCP LEASE  : 598, 600/300/525
MAC        : 00:50:79:66:68:7e
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

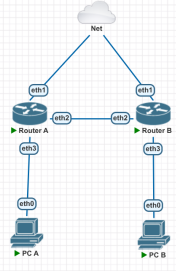
VPCS>
  
```

TERMINAL (4)

Physical 01 : 00 : 00 START

Lalu, lakukan yang serupa untuk PC B tapi ke Router B

8.

```

Terminal
Router A * Router B * PCA * PC B *
comment -- A descriptive comment for the rule
disable -- Disable items
edit --
enable -- Enable items
export -- Print or save an export script that can be used to restore configuration
find -- Find items by value
get -- Gets value of item's property
move -- Change order of items
print -- Print values of item properties
remove -- Remove item
reset-counters -- Set bytes and packets counters to 0
reset-counters-all -- Set all bytes and packets counters to 0
set -- Change item properties
unset --

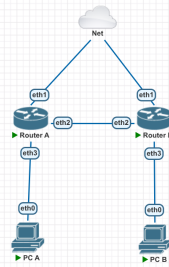
[admin@MikroTik] /ip> firewall nat remote 0
bad command name remote (line 1 column 14)
[admin@MikroTik] /ip> firewall nat remove 0
[admin@MikroTik] /ip> firewall nat print
Flags: X - disabled, I - invalid, D - dynamic
[admin@MikroTik] /ip> firewall nat add chain=srcnat out-interface=ether1 action=masquerade
[admin@MikroTik] /ip> ip route print
bad command name route (line 1 column 4)
[admin@MikroTik] /ip> route print
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
b - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 ADS 0.0.0.0/0 10.4.64.1 1
1 ADC 10.4.64.0/20 10.4.76.150 ether1 0
2 ADC 10.10.10.0/30 10.10.10.1 ether2 0
3 ADC 10.10.10.4/30 10.10.10.5 ether3 0
[admin@MikroTik] /ip>
  
```

TERMINAL (4)

Physical 01 : 00 : 00 START

Berikan konfigurasi NAT di Router A

9.



```

Terminal
Router A * Router B * PCA * PC B *
[admin@MikroTik] > ip address add
address: 10.10.10.9/30
interface: ether3
[admin@MikroTik] > ip dhcp-server setup
Select interface to run DHCP server on
dhcp server interface: ether3
Select network for DHCP addresses
dhcp address space: 10.10.10.8/30
Select gateway for given network
gateway for dhcp network: 10.10.10.9
Select pool of ip addresses given out by DHCP server
addresses to give out: 10.10.10.10
Select DNS servers
dns servers: 8.8.8.8
select lease time
Lease time: 10m
[admin@MikroTik] > ip firewall nat add chain=srcnat out-interface=ether1 action=masquerade
[admin@MikroTik] > ip route print
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS      PREF-SRC  GATEWAY      DISTANCE
0 AD  0.0.0.0/0        10.4.64.1    1
1 AD  10.4.64.0/20     10.4.76.151  0
2 AD  10.10.10.0/30    10.10.10.2   0
3 AD  10.10.10.8/30    10.10.10.9   0
[admin@MikroTik] >

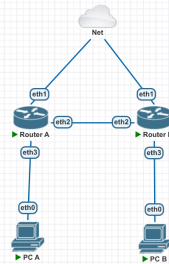
```

TERMINAL (4)

Physical 01 : 00 : 00 START

Berikan konfigurasi NAT di Router B

10.



```

Terminal
Router A * Router B * PCA * PC B *
RHOST:PORT : 127.0.0.1:30000
MTU : 1500
VPCS> ip dhcp
DORA IP 10.10.10.6/30 GW 10.10.10.5
VPCS> show ip
NAME : VPCS[1]
IP/MASK : 10.10.10.6/30
GATEWAY : 10.10.10.5
DNS : 8.8.8.8 202.46.129.3
DHCP SERVER : 10.10.10.5
DHCP LEASE : 598, 600/300/525
MAC : 00:50:79:66:68:7d
IP:PORT : 20000
RHOST:PORT : 127.0.0.1:30000
MTU : 1500
VPCS> ping 10.10.10.10
10.10.10.10 icmp_seq=1 timeout
VPCS> ping 8.8.8.8
8.8.8.8 icmp_seq=1 timeout
VPCS> ping 8.8.8.8
84 bytes from 8.8.8.8 icmp_seq=1 ttl=112 time=22.898 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=112 time=24.372 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=112 time=22.368 ms
VPCS>

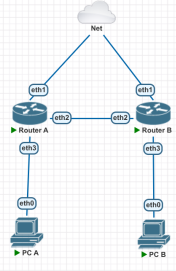
```

TERMINAL (4)

Physical 01 : 00 : 00 START

Lakukan percobaan ping ke DNS Google di internet dari PC A

11.

```

Terminal
Router A * Router B * PC A * PC B *
DNS :
MAC : 08:50:79:66:68:7e
LPORT : 20000
RHOST:PORT : 127.0.0.1:30000
MTU : 1500

VPCS> ip dhcp
DORA IP 10.10.10.10/30 GW 10.10.10.9

VPCS> show ip

NAME : VPCS[1]
IP/MASK : 10.10.10.10/30
GATEWAY : 10.10.10.9
DNS : 8.8.8.8 202.46.129.3
DHCP SERVER : 10.10.10.9
DHCP LEASE : 598, 600/300/525
MAC : 08:50:79:66:68:7e
LPORT : 20000
RHOST:PORT : 127.0.0.1:30000
MTU : 1500

VPCS> ping 8.8.8.8

8.8.8.8 icmp_seq=1 timeout
84 bytes from 8.8.8.8 icmp_seq=2 ttl=112 time=22.192 ms
PC
VPCS> ping 8.8.8.8

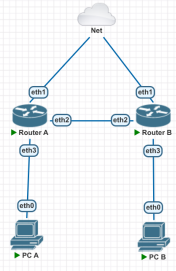
84 bytes from 8.8.8.8 icmp_seq=1 ttl=112 time=23.511 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=112 time=22.755 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=112 time=22.533 ms
PC
VPCS>
  
```

TERMINAL (4)

Physical 01 : 00 : 00 START

Lakukan percobaan ping ke DNS Google di internet dari PC B

12.

```

Terminal
Router A * Router B * PC A * PC B *
reset-counters-all -- Set all bytes and packets counters to 0
set -- Change item properties
unset --

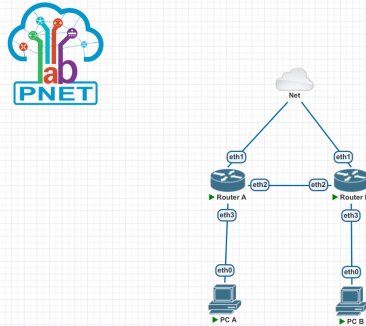
[admin@MikroTik] /ip> firewall nat remote 0
bad command name remote (line 1 column 14)
[admin@MikroTik] /ip> firewall nat remove 0
[admin@MikroTik] /ip> firewall nat print
Flags: X - disabled, I - invalid, D - dynamic
[admin@MikroTik] /ip> firewall nat add chain=srcnat out-interface=ether1 action=masquerade
[admin@MikroTik] /ip> ip route print
bad command name route (line 1 column 4)
[admin@MikroTik] /ip> route print
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 ADS 0.0.0.0/0 10.4.76.150 10.4.64.1 1
1 ADC 10.4.64.0/20 10.4.76.150 ether1 0
2 ADC 10.10.10.0/30 10.10.10.1 ether2 0
3 ADC 10.10.10.4/30 10.10.10.5 ether3 0
[admin@MikroTik] /ip> route add dst-address=10.10.10.8/30 gateway=10.10.10.2
[admin@MikroTik] /ip> route print
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 ADS 0.0.0.0/0 10.4.76.150 10.4.64.1 1
1 ADC 10.4.64.0/20 10.4.76.150 ether1 0
2 ADC 10.10.10.0/30 10.10.10.1 ether2 0
3 ADC 10.10.10.4/30 10.10.10.5 ether3 0
4 A S 10.10.10.8/30 10.10.10.2 1
[admin@MikroTik] /ip>
  
```

TERMINAL (4)

Physical 01 : 00 : 00 START

Lakukan routing statis di Router A agar PC A dapat terhubung ke PC B

13.

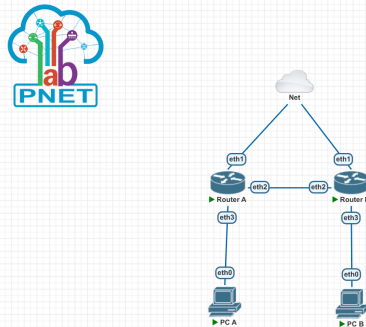


```

Terminal
Router A * Router B * PCA * PC B *
Select gateway for given network
gateway for dhcp network: 10.10.10.9
Select pool of ip addresses given out by DHCP server
addresses to give out: 10.10.10.10
Select DNS servers
dns servers: 8.8.8.8
Select lease time
lease time: 10m
[admin@MikroTik] > ip firewall nat add chain=srcnat out-interface=ether1 action=masquerade
[admin@MikroTik] > ip route print
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 ADS 0.0.0.0/0 10.4.76.151 10.4.64.1 1
1 ADC 10.4.64.0/20 10.10.10.2 ether1 0
2 ADC 10.10.10.0/30 10.10.10.2 ether2 0
3 ADC 10.10.10.8/30 10.10.10.9 ether3 0
[admin@MikroTik] > ip route add dst-address=10.10.10.4/30 gateway=10.10.10.1
[admin@MikroTik] > ip route print
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 ADS 0.0.0.0/0 10.4.76.151 10.4.64.1 1
1 ADC 10.4.64.0/20 10.10.10.2 ether1 0
2 ADC 10.10.10.0/30 10.10.10.2 ether2 0
3 A S 10.10.10.4/30 10.10.10.1 1
4 ADC 10.10.10.8/30 10.10.10.9 ether3 0
[admin@MikroTik] >
  
```

Lakukan routing statis di Router B agar PC B dapat terhubung ke PC A

14.



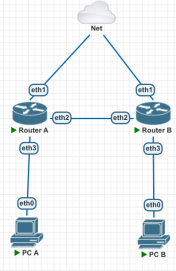
```

Terminal
Router A * Router B * PCA * PC B *
VPCS> show ip
NAME : VPCS[1]
IP/MASK : 10.10.10.6/30
GATEWAY : 10.10.10.5
DNS : 8.8.8.8 202.46.129.3
DHCP SERVER : 10.10.10.5
DHCP LEASE : 598, 600/300/525
MAC : 08:50:79:66:68:7d
LPORT : 20000
RHOST:PORT : 127.0.0.1:30000
MTU : 1500
VPCS> ping 10.10.10.10
10.10.10.10 icmp_seq=1 timeout
^C
VPCS> ping 8.8.8.8
8.8.8.8 icmp_seq=1 timeout
^C
VPCS> ping 8.8.8.8
84 bytes from 8.8.8.8 icmp_seq=1 ttl=112 time=22.898 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=112 time=24.372 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=112 time=22.388 ms
^C
VPCS> ping 10.10.10.10
84 bytes from 10.10.10.10 icmp_seq=1 ttl=62 time=5.007 ms
84 bytes from 10.10.10.10 icmp_seq=2 ttl=62 time=1.913 ms
84 bytes from 10.10.10.10 icmp_seq=3 ttl=62 time=3.075 ms
^C
VPCS>
  
```

Lakukan tes ping dari PC A ke PC B

15.





```

Terminal
Router A * Router B * PC A * PC B *
DORA IP 10.10.10.10/30 GW 10.10.10.9
VPCS> show ip
NAME      : VPCS[1]
IP/MASK    : 10.10.10.10/30
GATEWAY    : 10.10.10.9
DNS        : 8.8.8.8 202.46.129.3
DHCP SERVER : 10.10.10.9
DHCP LEASE  : 598, 600/300/525
MAC        : 00:50:79:66:68:7e
LPORT      : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500
VPCS> ping 8.8.8.8
8.8.8.8 icmp_seq=1 timeout
84 bytes from 8.8.8.8 icmp_seq=2 ttl=112 time=22.192 ms
VPCS> ping 8.8.8.8
84 bytes from 8.8.8.8 icmp_seq=1 ttl=112 time=23.511 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=112 time=22.755 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=112 time=22.533 ms
VPCS> ping 10.10.10.6
84 bytes from 10.10.10.6 icmp_seq=1 ttl=62 time=1.682 ms
84 bytes from 10.10.10.6 icmp_seq=2 ttl=62 time=1.511 ms
84 bytes from 10.10.10.6 icmp_seq=3 ttl=62 time=2.455 ms
84 bytes from 10.10.10.6 icmp_seq=4 ttl=62 time=1.268 ms
VPCS>
          
```

TERMINAL (4)
Physical 01 : 00 : 00 START

Kemudian, lakukan tes ping dari PC B ke PC A